ATR / Range Signal Analysis

Portfolio Builder

1 Objective

The objective of this test is to examine whether recent price volatility, measured using the Average True Range (ATR), contains predictive information about future stock returns.

A 20-day ATR is constructed from the True Range and normalized by the current closing price. The resulting signal is evaluated using cross-sectional decile analysis, Information Coefficient (IC), a long-short portfolio, and a HAC/Newey-West test.

2 Data

The dataset contains OHLCV data for 24 stocks.

- Valid observations: 34,656
- Number of stocks: 24
- Date range: 5 October 2016 to 5 August 2022

3 Signal Construction

The True Range is defined as

$$TR_t = \max\left(H_t - L_t,\, |H_t - C_{t-1}|,\, |L_t - C_{t-1}|\right).$$

The 20-day ATR is calculated using the previous 20 observations:

$$ATR_{20,t} = \frac{1}{20}\sum_{k=1}^{20} TR_{t-k}.$$

To make the measure comparable across stocks with different price levels, the ATR is normalized by the current closing price:

$$ATRSignal_t = \frac{ATR_{20,t}}{Close_t}.$$

The future 20-day return is defined as

$$R_{t,t+20} = \frac{Close_{t+20}}{Close_t} - 1.$$

4 Signal Statistics

The distribution of the normalized ATR signal was:

The average normalized ATR was approximately 2.72%, while the median was approximately 2.48%.

Statistic	Value
Mean	0.027165
Standard deviation	0.011645
Minimum	0.009831
25th percentile	0.020503
Median	0.024848
75th percentile	0.030605
Maximum	0.180225

Table 1: ATR Signal statistics

5 Decile Analysis

Stocks were sorted into ten cross-sectional groups each day according to their ATR signal.

Group 1 contains stocks with the lowest ATR values, while Group 10 contains stocks with the highest ATR values.

Group	Average Future 20-Day Return
1	1.7452%
2	1.6467%
3	1.4752%
4	1.3844%
5	1.3800%
6	1.5537%
7	1.6571%
8	1.4259%
9	1.7994%
10	2.4675%

Table 2: Future returns by ATR decile

The lowest ATR group produced an average future return of 1.7452%, while the highest ATR group produced 2.4675%.

The low-minus-high spread was therefore

$$1.7452\% - 2.4675\% = \boxed{-0.7223\%}.$$

Although the highest ATR group had the highest average future return, the decile pattern was not monotonic. The intermediate groups displayed irregular returns.

Therefore, the decile results do not provide strong evidence of a consistent relationship between ATR and future returns.

6 Information Coefficient

The daily cross-sectional Spearman Information Coefficient between the ATR signal and future 20-day returns was calculated.

The mean IC of 0.011550 is close to zero. The positive IC fraction is 50.07%, indicating that positive and negative daily relationships occurred with approximately equal frequency.

The p-value of 0.098831 is above the conventional 5% significance level. Therefore, the IC analysis does not provide statistically significant evidence of a consistent predictive relationship.

Statistic	Value
Mean IC	0.011550
Median IC	0.000870
IC standard deviation	0.265741
Positive IC fraction	50.07%
t-statistic	1.651611
p-value	0.098831

Table 3: ATR Information Coefficient results

7 Long-Short Portfolio

A long-short portfolio was constructed using the extreme ATR deciles:

- Long position: lowest ATR decile
- Short position: highest ATR decile

The long-short return is

$$R_{LS} = R_{G1} - R_{G10}.$$

The portfolio statistics were:

Statistic	Value
Mean return	-0.007223
Median return	-0.002634
Standard deviation	0.067956
Annualized return	-182.0220%
Annualized volatility	107.8767%
Sharpe-like ratio	-1.6873
Ordinary t-statistic	-4.039055
Ordinary p-value	0.000056

Table 4: ATR long-short portfolio

The low-minus-high portfolio produced a negative mean return of 0.7223%.

The ordinary t-test suggests statistical significance. However, the 20-day forward returns overlap across consecutive observations, so the ordinary standard error may understate the true uncertainty.

The annualized return should therefore not be interpreted as an expected annual portfolio return.

8 HAC / Newey-West Test

To account for autocorrelation caused by overlapping 20-day forward returns, a HAC/Newey-West test with 20 lags was performed.

The HAC p-value is 0.194252, which is well above the conventional 5% significance level.

Therefore, the long-short portfolio does not provide statistically significant evidence of a return difference after accounting for autocorrelation and overlapping observations.

Statistic	Value
Mean return	-0.007223
HAC standard error	0.005564
HAC t-statistic	-1.298103
HAC p-value	0.194252

Table 5: HAC/Newey-West results

9 Interpretation

The ATR signal does not show a clear monotonic relationship with future 20-day returns.

Although the highest ATR decile produced the highest average future return, the intermediate deciles did not follow a consistent pattern. The mean IC of 0.011550 is also close to zero.

The ordinary long-short test initially appears significant, but this significance disappears after applying the HAC/Newey-West correction.

Therefore, the evidence does not support ATR as a reliable standalone return-prediction factor.

Instead, ATR is more appropriately viewed as a measure of recent price risk and volatility. It may be useful for portfolio construction, risk management, position sizing, or filtering rather than as a primary alpha signal.

10 Conclusion

The ATR / Range analysis provides no statistically significant standalone predictive relationship with 20-day future returns after accounting for overlapping observations.

The main findings are:

- Mean IC: 0.011550
- IC p-value: 0.098831
- Low-minus-high spread: -0.7223%
- Ordinary long-short p-value: 0.000056
- HAC/Newey-West p-value: 0.194252

The HAC result is the more appropriate inference because the 20-day forward returns overlap.

Therefore, ATR is classified as a

Risk / Portfolio Construction Filter

rather than a standalone alpha factor.

It can potentially be used later for volatility control, position sizing, or as a conditioning variable in the multi-factor portfolio model.